Suhas Sudhir Bharadwaj, Ph.D.

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Education

Ph.D. in Industrial and Systems Engineering at Auburn University, Auburn, AL *Jan 2020 - Aug 2025*

Accomplishments

- Data-driven Public Health professional with 5 years of advanced training in Occupational Health and Statistics, applying epidemiologic methods, quantitative analysis, and evidence-based approaches to investigate health risks, evaluate interventions, and develop actionable solutions for complex public health challenges.
- Recognized for excellence in occupational and public health innovation through competitive awards from leading organizations including the American Society of Safety Professionals (ASSP) and National Safety Council (NSC), totaling ~$22,000, demonstrated community engagement through fundraising efforts supporting the National Multiple Sclerosis Society (NMSS).
- Experienced cross-functional collaborator with demonstrated ability to independently execute research and applied public health projects, including 5 research studies, 10 practice-based projects, 25 graduate credit hours in Statistics, and authorship of 2 peer-reviewed publications in high-impact journals.

Work Experience

Public Health Data Scientist and Mentor

Doctoral Student at Auburn University, Auburn, AL *Jan 2020 - Aug 2025*

- Analyzed longitudinal healthcare datasets containing 2.5 million records, 15K entities, and geographical clustering using mixed-effects models to evaluate nursing home quality measures and COVID-19 outcomes, translating statistical findings into evidence supporting healthcare quality improvement.
- Performed large-scale analysis of public health datasets using SAS and R in a high-performance computing (HPC) environment to identify health trends and support occupational health and injury prevention research.
- Conducted longitudinal and compositional data analyses using parametric and non-parametric statistical methods across 135 datasets, 32 participants, and 50+ variables, applying analytical reasoning to support data-driven research findings.
- Showcased collaborative planning and research logistics skills by coordinating approximately 60 field data collection visits, collecting data from 32 manufacturing workers, and facilitating execution by a 5-member team on an NIH-funded research initiative, ensuring efficient and high-quality data collection for downstream analysis.
- Demonstrated strong interpersonal and mentoring skills in engineering education and technical instruction, mentoring 500+ students over 13 semesters, translating complex engineering, programming, statistical, and analytical concepts into accessible instruction and supporting data-driven problem solving.
- Exhibited teaching and communication skills supported by hands-on engineering laboratory instruction, conducting 50+ lab sessions and 2 course lectures to improve student engagement and practical application of course material.
- Optimized student learning and course delivery for 40+ students each semester by streamlining Canvas LMS content management, conducting office hours, providing individualized feedback, and analyzing assignment

performance, ensuring 100% on-time access to instructional materials and supporting improved learning outcomes.

Research Investigator and Project Leader

Graduate Student at University of Minnesota - Twin Cities, Minneapolis, MN *Sept 2015 - Dec 2017*

- Collected, managed, and analyzed behavioral and observational data to evaluate injury prevention interventions and generate evidence supporting transportation safety initiatives.
- Designed and conducted human-subject research, coordinating laboratory and field data collection to generate high-quality physiological and kinematic datasets for statistical analysis, interpretation, and dissemination of research findings.

Additional Experience

Resourcefulness and Leadership in OSH

EHS Specialist at J. M. Huber Corporation, Quincy, FL *Mar 2025 - Jul 2026*

- Launched HSMS' LSR initiative, closing all identified compliance gaps for significantly improving safety culture while reducing incident risk as well as site-level plans to achieve 2026 HESS objectives through engaging employees at all levels, building cross-functional teams, and structured action tracking at 2 sites.
- Implemented silica sampling programs across 2 sites, collecting 24 samples according to modified NIOSH 0600/7500 methods and analysing according to modified OSHA ID-142 Gravimetry/X-ray diffraction, resulting in sustained regulatory compliance through silica exposure below 25 µg/m3 (8-hr TWA), improving indoor air quality for 43 employees.
- Established Safety Committees at 2 sites with 20 engaged employees, conducting 20 meetings that generated 10 actionable ideas for health and safety improvements planned for 2026.

Skills

- Statistical Analysis: Regression, Mixed-Effects and Longitudinal Analysis, Statistical and Predictive Modeling
- Data Management and Reproducible Research: Data Cleaning, Data Wrangling, Data Quality Assurance, Version Control, Workflow Automation, Dynamic Reporting
- Data Visualization and Scientific Communication: Data Storytelling, Scientific Visualization, Dashboards, Technical Reports, Presentation Briefs, Research Summaries
- Geospatial: Spatial Analysis and Modeling
- Programming and Computational Methods: R, Python, SAS, SQL
- Machine Learning: Feature Engineering, Model Testing, Development and Validation
- Public Health and Research Methods: Experimental and Observational Studies, Program Evaluation, Stakeholder Engagement, Evidence-Based Interventions

Publications

- Ali, H., & **Bharadwaj, S. S.** (2026). Demystifying quality metrics and unveiling the true measure of quality of care in nursing homes. JMIR Human Factors, 13, e72770. https://doi.org/10.2196/72770
- Ali, H., Ali, D., Fatemi, Y., & **Bharadwaj, S. S.** (2025). ChatGPT perceptions, experiences, and uses with emphasis

on academia. Frontiers in Education, 10. https://doi.org/10.3389/feduc.2025.1559509

Authorized to work in the U.S. without sponsorship now or in the future